MATH1025B
Essential Mathematical Methods

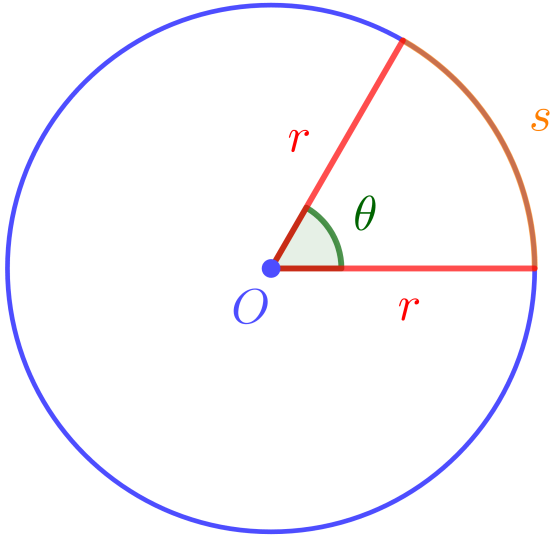
Chapter 1: Trigonometry

September 9, 2025

Definition 1.1 (Radian Measure)

The magnitude θ in radian of an angle subtended at the centre of a circle is the ratio of the arc length s to the radius r of the circle, that is,

$$\theta = \frac{s}{r}.$$



Angles

If the circle has radius $r = 1$, then the central angle θ measured in radians is just the length of the arc that the angles cuts from the unit circle. Since one complete revolution of the unit circle is 360° or 2π radians, we have

$$\pi \text{ radian} = 180^\circ$$

and thus

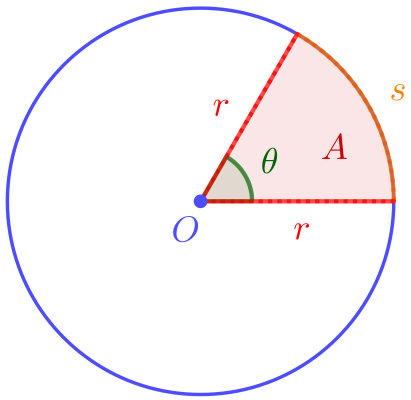
$$1 \text{ radian} = \frac{180}{\pi}(\approx 57.3) \text{ degrees} \qquad \text{or} \qquad 1 \text{ degree} = \frac{\pi}{180}(\approx 0.017) \text{ radians.}$$

Angles measured in degrees and radians

Degrees	0	30	45	60	90	120	135	150	180	270	360
θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π	$\frac{3\pi}{2}$	2π

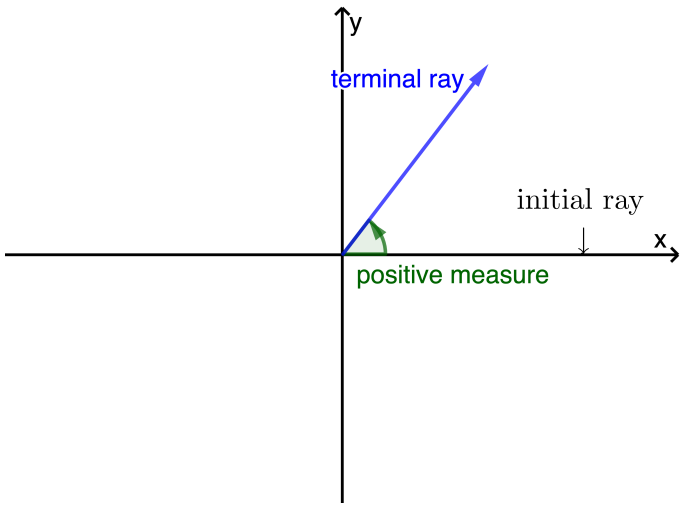
Area of Sector and Arclength

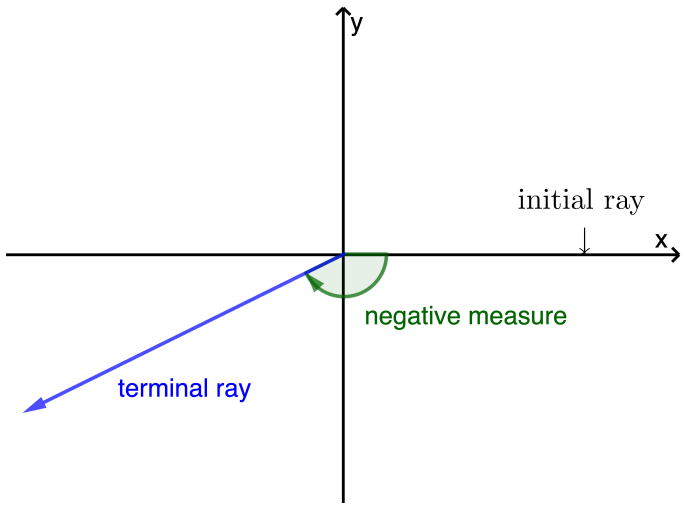
$$A = \frac{1}{2}r^2\theta, \qquad s = r\theta.$$



Angles in Standard Position

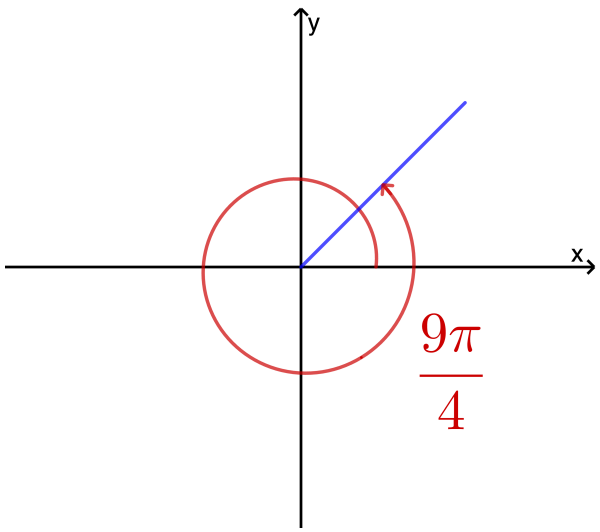
An angle in the xy-plane is said to be in **standard position** if its vertex lies at the origin and its initial ray lies along the positive x-axis. Angles measured counterclockwise from the positive x-axis are assigned positive measures; angles measured clockwise are assigned negative measures.

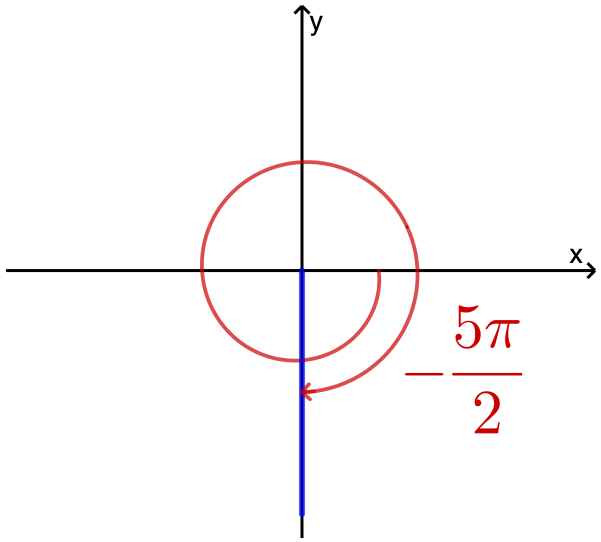




Angles

Angles describing counterclockwise rotations can go beyond 2π radians or $360°$. Similarly, angles describing clockwise rotations can have negative measures of all sizes.





Angle Convention: Use Radians

From now on, all angles are measured in radians unless stated otherwise. When we talk about the angle $\pi/3$, we mean $\pi/3$ radians (which is $60°$), not $\pi/3$ degrees. Very often we will drop the unit and simply say $\pi/3$.

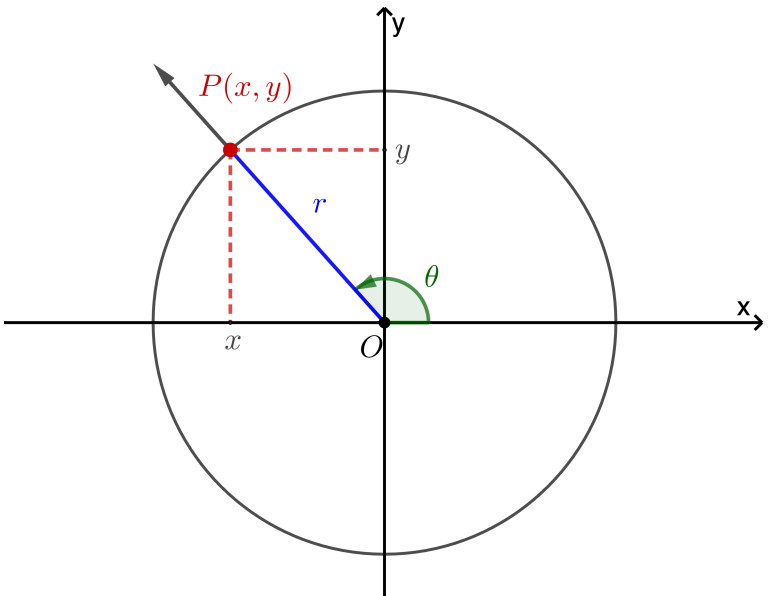
Definition 1.2 (Trigonometric Functions)

Place the angle θ in standard position in a circle of radius r. The angle's terminal ray intersect the circle at the point $P(x, y)$. The trigonometric functions are then defined as follows: (Note: $r = \sqrt{x^2 + y^2}$.)

$$\textbf{sine:} \quad \sin\theta = \frac{y}{r} \qquad \textbf{cosecant:} \quad \csc\theta = \frac{r}{y}$$

$$\textbf{cosine:} \quad \cos\theta = \frac{x}{r} \qquad \textbf{secant:} \quad \sec\theta = \frac{r}{x}$$

$$\textbf{tangent:} \quad \tan\theta = \frac{y}{x} \qquad \textbf{cotangent:} \quad \cot\theta = \frac{x}{y}$$



Trigonometric Functions

Values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ for selected values of θ

Degrees	0	30	45	60	90
θ (radians)	0	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$
$\sin\theta$	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1
$\cos\theta$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0
$\tan\theta$	0	$\frac{\sqrt{3}}{3}$	1	$\sqrt{3}$	undefined

Basic transformation of $\sin\theta$, $\cos\theta$ and $\tan\theta$

	$-\theta$	$\pi-\theta$	$\pi+\theta$	$2\pi+\theta$	$\frac{\pi}{2}-\theta$	$\frac{\pi}{2}+\theta$
sin	$-\sin\theta$	$\sin\theta$	$-\sin\theta$	$\sin\theta$	$\cos\theta$	$\cos\theta$
cos	$\cos\theta$	$-\cos\theta$	$-\cos\theta$	$\cos\theta$	$\sin\theta$	$-\sin\theta$
tan	$-\tan\theta$	$-\tan\theta$	$\tan\theta$	$\tan\theta$	$\cot\theta$	$-\cot\theta$

Basic Trigonometric Identities

Theorem 1.1 (Periodicity of Trigonometric Functions)

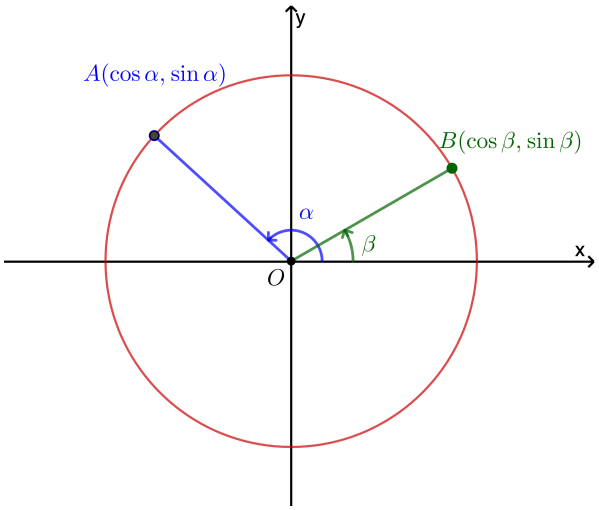
$$\begin{aligned} \sin(\theta + 2\pi) &= \sin\theta, & \cos(\theta + 2\pi) &= \cos\theta, \\ \sec(\theta + 2\pi) &= \sec\theta, & \csc(\theta + 2\pi) &= \csc\theta, \\ \tan(\theta + \pi) &= \tan\theta, & \cot(\theta + \pi) &= \cot\theta. \end{aligned}$$

Theorem 1.2 (Pythagorean Theorem)

- $\sin^2\theta + \cos^2\theta = 1$
- $1 + \tan^2\theta = \sec^2\theta$
- $1 + \cot^2\theta = \csc^2\theta$

Compound Angle Formulas

Consider two points $A(\cos\alpha, \sin\alpha)$ and $B(\cos\beta, \sin\beta)$ on the unit circle.



By the Cosine Formula,

$$\begin{aligned} AB^2 &= OA^2 + OB^2 - 2(OA)(OB)\cos\angle AOB \\ &= 2 - 2\cos(\alpha - \beta). \end{aligned}$$

On the other hand, the Distance formula implies that

$$\begin{aligned} AB^2 &= (\cos\alpha - \cos\beta)^2 + (\sin\alpha - \sin\beta)^2 \\ &= 2 - 2\cos\alpha\cos\beta - 2\sin\alpha\sin\beta. \end{aligned}$$

Hence

$$\cos(\alpha - \beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta. \quad (*)$$

Theorem 1.3 (Compound Angle Formulas)

- $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
- $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
- $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$
- $\tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$

Proof. Substitute α by $\frac{\pi}{2} - \alpha$ and β by $-\beta$ in $(*)$, we have

$$\begin{aligned}\sin(\alpha-\beta) &= \cos(\frac{\pi}{2} - \alpha + \beta) = \cos(\frac{\pi}{2} - \alpha)\cos(-\beta) + \sin(\frac{\pi}{2} - \alpha)\sin(-\beta)\\ &= \sin\alpha\cos\beta - \cos\alpha\sin\beta.\end{aligned}$$

Taking quotient of the above formulas, we have

$$\tan(\alpha-\beta) = \frac{\sin(\alpha-\beta)}{\cos(\alpha-\beta)} = \frac{\sin\alpha\cos\beta - \cos\alpha\sin\beta}{\cos\alpha\cos\beta + \sin\alpha\sin\beta} = \frac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}.$$

The formulas for $\sin(\alpha+\beta), \cos(\alpha+\beta), \tan(\alpha+\beta)$ can be obtained by replacing β with $-\beta$.

Theorem 1.4 (Compound Angle Formulas)

- $\sin(\alpha+\beta) = \sin\alpha\cos\beta + \cos\alpha\sin\beta$
- $\cos(\alpha+\beta) = \cos\alpha\cos\beta - \sin\alpha\sin\beta$
- $\tan(\alpha+\beta) = \dfrac{\tan\alpha + \tan\beta}{1 - \tan\alpha\tan\beta}$
- $\sin(\alpha-\beta) = \sin\alpha\cos\beta - \cos\alpha\sin\beta$
- $\cos(\alpha-\beta) = \cos\alpha\cos\beta + \sin\alpha\sin\beta$
- $\tan(\alpha-\beta) = \dfrac{\tan\alpha - \tan\beta}{1 + \tan\alpha\tan\beta}$

The following results can be deduced easily from the Compound Angle Formulas.

Theorem 1.5 (Double Angle Formulas)

- $\sin(2\alpha) = 2\sin\alpha\cos\alpha$
- $\cos(2\alpha) = \cos^2\alpha - \sin^2\alpha = 2\cos^2\alpha - 1 = 1 - 2\sin^2\alpha$
- $\tan(2\alpha) = \dfrac{2\tan\alpha}{1 - \tan^2\alpha}$

Theorem 1.6 (Product to Sum Formulas)

- $\sin\alpha\cos\beta = \dfrac{1}{2}\left(\sin(\alpha+\beta)+\sin(\alpha-\beta)\right)$
- $\cos\alpha\sin\beta = \dfrac{1}{2}\left(\sin(\alpha+\beta)-\sin(\alpha-\beta)\right)$
- $\sin\alpha\sin\beta = -\dfrac{1}{2}\left(\cos(\alpha+\beta)-\cos(\alpha-\beta)\right)$
- $\cos\alpha\cos\beta = \dfrac{1}{2}\left(\cos(\alpha+\beta)+\cos(\alpha-\beta)\right)$

Theorem 1.7 (Sum to Product Formulas)

- $\sin A + \sin B = 2\sin\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$
- $\sin A - \sin B = 2\sin\left(\frac{A-B}{2}\right)\cos\left(\frac{A+B}{2}\right)$
- $\cos A + \cos B = 2\cos\left(\frac{A+B}{2}\right)\cos\left(\frac{A-B}{2}\right)$
- $\cos A - \cos B = -2\sin\left(\frac{A+B}{2}\right)\sin\left(\frac{A-B}{2}\right)$

Example 1.1

Express $\sin 15^\circ$ in surd form.

$$\begin{aligned}
\sin 15^\circ &= \sin(45^\circ - 30^\circ) \\
&= \sin 45^\circ \cos 30^\circ - \sin 30^\circ \cos 45^\circ \\
&= \frac{\sqrt{2}}{2} \cdot \frac{\sqrt{3}}{2} - \frac{1}{2} \cdot \frac{\sqrt{2}}{2} \\
&= \frac{1}{4}\left(\sqrt{6} - \sqrt{2}\right).
\end{aligned}$$

Example 1.2

In $\triangle ABC$, prove that $\tan A + \tan B + \tan C = \tan A \tan B \tan C$.

Note that $A + B + C = \pi$. So

$$\tan C = \tan(\pi - A - B) = -\tan(A + B) = -\frac{\tan A + \tan B}{1 - \tan A \tan B}.$$

Therefore,

$$\begin{aligned}
\text{L.H.S.} &= \tan A + \tan B + \tan C \\
&= \tan A + \tan B - \frac{\tan A + \tan B}{1 - \tan A \tan B} \\
&= -\frac{\tan A + \tan B}{1 - \tan A \tan B} \cdot \tan A \tan B \\
&= \tan A \tan B \tan C \\
&= \text{R.H.S.}
\end{aligned}$$

Example 1.3 (t-substitution)

Let $t = \tan\dfrac{\theta}{2}$. Express $\sin\theta$, $\cos\theta$, $\tan\theta$ in terms of t.

$$\begin{aligned}
\sin\theta &= \sin(2\cdot\frac{\theta}{2}) = 2\sin\frac{\theta}{2}\cos\frac{\theta}{2} \\
&= 2\frac{\sin\frac{\theta}{2}}{\cos\frac{\theta}{2}}\cos^2\frac{\theta}{2} = \frac{2\tan\frac{\theta}{2}}{\sec^2\frac{\theta}{2}} \\
&= \frac{2\tan\frac{\theta}{2}}{1+\tan^2\frac{\theta}{2}} = \frac{2t}{1+t^2}.
\end{aligned}$$

Similarly, we can show that $\cos\theta = \dfrac{1-t^2}{1+t^2}$ and $\tan\theta = \dfrac{2t}{1-t^2}$.

Basic Trigonometric Identities

Exercise 1.1

Prove the following identities:

(a) $\tan(A+B)+\tan(A-B)=\dfrac{2\sin 2A}{\cos 2A+\cos 2B}$

(b) (Triple angle formulas) $\begin{cases}\sin 3\theta = 3\sin\theta - 4\sin^3\theta \\ \cos 3\theta = 4\cos^3\theta - 3\cos\theta\end{cases}$

Exercise 1.2

Prove the following identities:

(a) $4\cos\theta\cos(\frac{2\pi}{3}+\theta)\cos(\frac{2\pi}{3}-\theta)=\cos 3\theta$

(b) $\sin^2\theta+\sin^2\varphi-\sin^2(\theta-\varphi)=2\sin\theta\sin\varphi\cos(\theta-\varphi)$

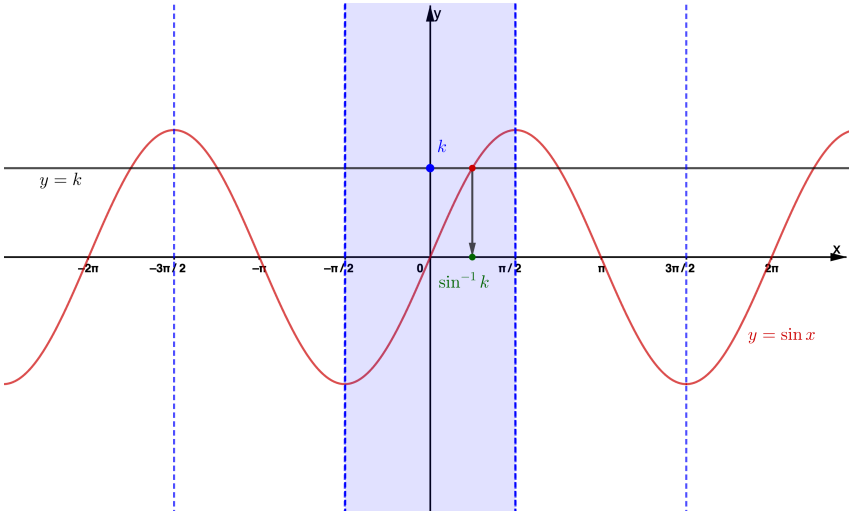
Exercise 1.3

In $\triangle ABC$, prove that

(a) $\tan\dfrac{A}{2}\tan\dfrac{B}{2}+\tan\dfrac{B}{2}\tan\dfrac{C}{2}+\tan\dfrac{C}{2}\tan\dfrac{A}{2}=1$

(b) $\cos^2 A+\cos^2 B+\cos^2 C=1-2\cos A\cos B\cos C$

Sine function



Suppose $k \in [-1, 1]$. Then there is a unique number $\alpha \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, denoted by $\sin^{-1} k$, such that

$$\sin \alpha = k.$$

Moreover

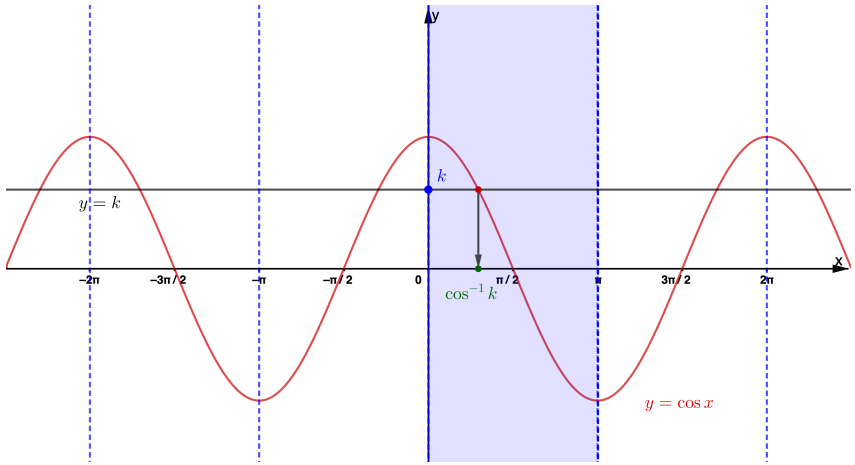
$$\sin(\theta) = \sin(\pi - \theta) = \sin(\theta + 2\pi) \quad \text{for any } \theta \in \mathbb{R}.$$

Theorem 1.8 (General solution formula: sine)

Suppose $\sin \theta = k$, *where* $k \in [-1, 1]$. *Then*

$$\theta = \alpha + 2n\pi \quad \text{or} \quad \theta = \pi - \alpha + 2n\pi, \quad \text{where } n \in \mathbb{Z} \text{ and } \alpha = \sin^{-1} k.$$

Cosine function



Suppose $k \in [-1, 1]$. Then there is a unique number $\alpha \in [0, \pi]$, denoted by $\cos^{-1} k$, such that

$$\cos \alpha = k.$$

Moreover

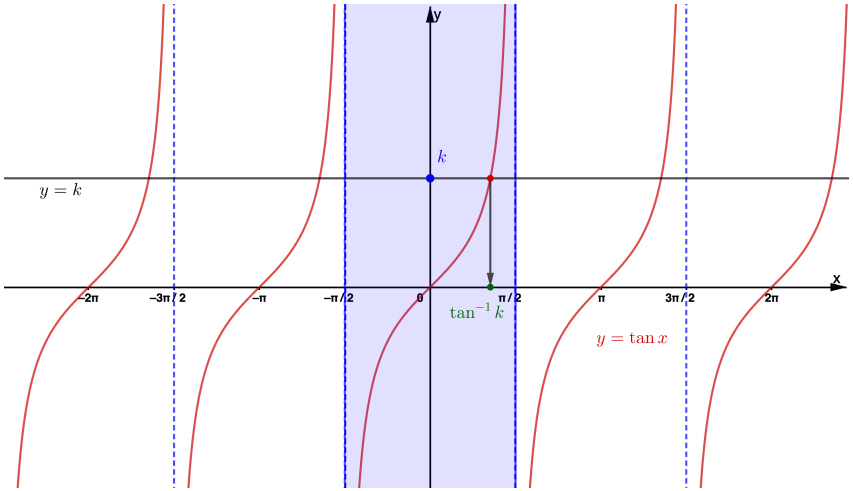
$$\cos(\theta) = \cos(-\theta) = \cos(\theta + 2\pi) \quad \text{for any } \theta \in \mathbb{R}.$$

Theorem 1.9 (General solution formula: cosine)

Suppose $\cos \theta = k$, *where* $k \in [-1, 1]$. *Then*

$$\theta = \pm\alpha + 2n\pi, \quad \textit{where } n \in \mathbb{Z} \textit{ and } \alpha = \cos^{-1} k.$$

Tangent function



Suppose $k \in \mathbb{R}$. Then there is a unique number $\alpha \in (-\frac{\pi}{2}, \frac{\pi}{2})$, denoted by $\tan^{-1} k$, such that

$$\tan \alpha = k.$$

Moreover

$$\tan(\theta) = \tan(\theta + \pi) \quad \text{for any } \theta \in \mathbb{R} \backslash \{m\pi + \frac{\pi}{2} : m \in \mathbb{Z}\}.$$

Theorem 1.10 (General solution formula: tangent)

Suppose $\tan \theta = k$, *where* $k \in \mathbb{R}$. *Then*

$$\theta = \alpha + n\pi, \quad \textit{where } n \in \mathbb{Z} \textit{ and } \alpha = \tan^{-1} k.$$

Example 1.4

Find the general solution of $\dfrac{1+\tan x}{1-\tan x} = 1 + \sin 2x$.

Let $t = \tan x$. Then $\sin 2x = \frac{2\tan x}{1+\tan^2 x} = \frac{2t}{1+t^2}$. So

$$\begin{aligned} \frac{1+t}{1-t} &= 1 + \frac{2t}{1+t^2} \\ (1+t)(1+t^2) &= (1-t)(1+t^2) + 2t(1-t) \\ 2t^2(1+t) &= 0. \end{aligned}$$

Hence,

$$\begin{aligned} \tan x = 0 \quad &\text{or} \quad \tan x = -1 \\ x = n\pi \quad &\text{or} \quad x = -\frac{\pi}{4} + n\pi, \qquad \text{where } n \in \mathbb{Z}. \end{aligned}$$

Example 1.5

(a) Show that $\cos 2x + 2\cos 3x + \cos 4x = 4\cos^2 \frac{x}{2} \cos 3x$.

(b) Hence, solve for all solutions in $[0, 2\pi]$ of the equation $\cos 2x + 2\cos 3x + \cos 4x = 0$.

(a) By Sum to Product Formula then Double Angle Formula,

$$\begin{aligned}\cos 2x + \cos 4x + 2\cos 3x &= 2\cos(3x)\cos(-x) + 2\cos 3x \\ &= 2\cos 3x\,(\cos x + 1) \\ &= 4\cos 3x \cos^2 \frac{x}{2}.\end{aligned}$$

(b) By (a), we have

$$\begin{aligned} \cos 3x = 0 \quad &\text{or} \quad \cos\frac{x}{2} = 0 \\ 3x = \frac{\pi}{2} + n\pi \quad &\text{or} \quad \frac{x}{2} = \pm\frac{\pi}{2} + 2n\pi, \qquad \text{where } n \in \mathbb{Z} \\ x = \frac{\pi}{6} + \frac{n\pi}{3} \quad &\text{or} \quad x = \pi + 2n\pi, \qquad \text{where } n \in \mathbb{Z}. \end{aligned}$$

Hence, the solutions in $[0, 2\pi]$ are $\frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \frac{7\pi}{6}, \frac{3\pi}{2}, \frac{11\pi}{6}$ and π.

Subsidiary Angle Form

Goal:

Express $a\sin\theta + b\cos\theta$ in the form of

$$R\sin(\theta+\alpha) \quad \text{or} \quad R\cos(\theta-\alpha), \quad \text{where } R>0 \text{ and } 0 \le \alpha < 2\pi.$$

Suppose $a \neq 0$ and

$$a\sin\theta + b\cos\theta = R\sin(\theta+\alpha) = R\sin\theta\cos\alpha + R\cos\theta\sin\alpha.$$

Want: find $R>0$ and $\alpha \in [0, 2\pi)$ such that $\begin{cases} R\cos\alpha = a & (1) \\ R\sin\alpha = b & (2) \end{cases}$.

$(1)^2 + (2)^2$: $\quad R^2 = a^2 + b^2 \implies R = \sqrt{a^2+b^2}$

$(2)/(1)$: $\quad \tan\alpha = \dfrac{b}{a}$ (Need to find α that satisfies both (1) and (2).)

Hence $a\sin\theta + b\cos\theta = \sqrt{a^2+b^2}\sin(\theta+\alpha)$, where $\tan\alpha = \frac{b}{a}$.

Exercise 1.4

Let $b \neq 0$. Show that $a\sin\theta + b\cos\theta = \sqrt{a^2+b^2}\cos(\theta-\alpha)$, where $\tan\alpha = \frac{a}{b}$.

Example 1.6

Express $\sqrt{3}\sin\theta - \cos\theta$ in the form of $R\sin(\theta+\alpha)$ and $R\cos(\theta-\alpha)$, where $R > 0$ and $0 \le \alpha < 2\pi$.

Write $\sqrt{3}\sin\theta - \cos\theta = R\sin(\theta+\alpha) = R\sin\theta\cos\alpha + R\cos\theta\sin\alpha$. Then

$$\begin{cases} R\cos\alpha = \sqrt{3} & (1) \\ R\sin\alpha = -1 & (2) \end{cases} \implies \cos\alpha > 0 \text{ and } \sin\alpha < 0 \implies \frac{3\pi}{2} < \alpha < 2\pi.$$

$$(1)^2 + (2)^2: \qquad R^2 = 4 \implies R = 2$$

$$(2)/(1): \qquad \tan\alpha = -\frac{1}{\sqrt{3}} \implies \alpha = \frac{5\pi}{6} \text{ (rejected)} \quad \text{or} \quad \alpha = \frac{11\pi}{6}$$

Hence $\sqrt{3}\sin\theta - \cos\theta = 2\sin(\theta + \frac{11\pi}{6})$.

Example 1.6

Express $\sqrt{3}\sin\theta - \cos\theta$ in the form of $R\sin(\theta+\alpha)$ and $R\cos(\theta-\alpha)$, where $R > 0$ and $0 \le \alpha < 2\pi$.

Similarly, $\sqrt{3}\sin\theta - \cos\theta = R\cos(\theta-\alpha) = R\cos\theta\cos\alpha + R\sin\theta\sin\alpha$. Then

$$\begin{cases} R\cos\alpha = -1 & (1) \\ R\sin\alpha = \sqrt{3} & (2) \end{cases} \implies \cos\alpha < 0 \text{ and } \sin\alpha > 0 \implies \frac{\pi}{2} < \alpha < \pi.$$

$$(1)^2 + (2)^2: \qquad R^2 = 4 \implies R = 2$$

$$(2)/(1): \qquad \tan\alpha = -\sqrt{3} \implies \alpha = \frac{2\pi}{3} \quad \text{or} \quad \alpha = \frac{5\pi}{3} \text{ (rejected)}$$

Hence $\sqrt{3}\sin\theta - \cos\theta = 2\cos(\theta - \frac{2\pi}{3})$.

Example 1.7

Solve the equation $8\cos x = \dfrac{1}{\cos x} - \dfrac{\sqrt{3}}{\sin x}$.

Multiplying both sides by $\sin x \cos x$, we have

$$\begin{aligned}
8\cos^2 \sin x &= \sin x - \sqrt{3}\cos x \\
2\sin 2x \cos x &= \frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x \\
\sin 3x + \sin x &= \frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x \\
\sin 3x &= -\frac{1}{2}\sin x - \frac{\sqrt{3}}{2}\cos x \\
\sin 3x &= \sin(x + \frac{4\pi}{3}).
\end{aligned}$$

$$\therefore\ 3x = x + \frac{4\pi}{3} + 2n\pi \quad \text{or} \quad 3x = \pi - \left(x + \frac{4\pi}{3}\right) + 2n\pi, \qquad \text{where } n \in \mathbb{Z}$$

$$\text{i.e.}\quad x = \frac{2\pi}{3} + n\pi \quad \text{or} \quad x = -\frac{\pi}{12} + \frac{n\pi}{2}, \qquad \text{where } n \in \mathbb{Z}$$

Since $\sin x \cos x \neq 0$, these are the solutions of the given equation.